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Lagrangian Relaxation and the Single-Source Capacitated Facility-Location Problem

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Lagrangian relaxation has been widely used in solving a number of hard combinatorial optimization problems. The success of the approach depends on the structure of the problem and on the values assigned to the Lagrange multipliers. A recent paper on the single-source capacitated facility-location problem proposed the use of Lagrangian relaxation in which the capacity constraints were relaxed. In this paper, a class of such problems is defined for which the proposed relaxation is guaranteed to result in an infeasible solution, irrespective of the values assigned to the Lagrange multipliers. In these cases, the bounds on the optimal solution, obtained from the relaxation, are generally poor. It is concluded that, when using Lagrangian relaxation, it may be worthwhile carrying out a preliminary analysis to determine the potential viability of the approach before extensive development takes place.

Key words: combinatorial optimization, Lagrangian relaxation, location

INTRODUCTION

Lagrangian relaxation has been extensively applied to many combinatorial optimization problems during the last 10–15 years. Applications include production scheduling,^{1–3} assembly system design,^{4,5} hierarchical production planning,⁶ vehicle routing and scheduling,^{7–10} personnel scheduling,¹¹ manpower planning,¹² capital budgeting,¹³ data-base condensation¹⁴ and concentrator location in computer communication networks.¹⁵ The term ‘Lagrangian relaxation’ was first coined by Geoffrion¹⁶ in 1974 and, as Fisher¹⁷ states in an excellent review of the use of Lagrangian relaxation in combinatorial optimization, is ‘an important new computational technique in the management scientist’s arsenal’.

Given a combinatorial optimization problem expressed as an integer programme, the constraints are partitioned into two sets. A relaxed problem is constructed in which one set of constraints is removed (relaxed), and an augmented objective function is formed which incorporates penalties for violations of the removed constraints. The penalties are controlled by a set of Lagrange multipliers. The idea is that the relaxed problem, for a specified set of multipliers, is easy (sometimes trivial) to solve. The aim is to obtain a good feasible solution and/or a good bound on the optimal solution to the unrelaxed problem. The degree to which this aim is achieved depends on the structure of the problem and also on the values specified for the Lagrange multipliers. The approach involves solving the relaxed problem with different sets of multipliers. The procedure used for updating the multipliers is usually a version of the subgradient method.

The single-source capacitated facility-location problem (defined in the next section) is a well known, NP-hard combinatorial optimization problem. Several authors have studied the problem. Neebe and Rao¹⁸ formulate the problem as a set-partitioning problem and use branch and bound. Barceló and Casanovas¹⁹ (and subsequent comment by Sridharan²⁰) propose a Lagrangian relaxation in which the assignment constraints are relaxed. Klinecicz and Luss²¹ also propose Lagrangian relaxation but with the capacity constraints relaxed. Their approach is supplemented by a heuristic procedure (called ADD), which determines an initial upper bound and feasible solution, and a final adjustment procedure, which looks for simple reassignments of customers to open facilities, resulting in a cost reduction. They considered a set of 24 test problems and reported that their overall approach performed well. Klinecicz and Luss recognized that the proposed Lagrangian relaxation may not yield a feasible solution. Indeed, they provide a small illustrative example of such a case and, in one of the 24 test problems, the Lagrangian relaxation failed. However, the ADD procedure does guarantee that a feasible, although in some cases poor, solution is obtained in all cases.

The purposes of this paper is to define a class of single-source capacitated facility-location problems for which Lagrangian relaxation with the capacity constraints relaxed is guaranteed to produce infeasible solutions to the unrelaxed problems, irrespective of the values assigned to the Lagrange multipliers. In such cases the lower bound on the optimal solution to the unrelaxed problem, obtained from the solution to the relaxed problem, will generally be poor. The class is determined by the relationship between the fixed costs and the variable costs associated with the facilities.

THE SINGLE-SOURCE CAPACITATED FACILITY-LOCATION PROBLEM

Let $M = \{1, 2, \dots, m\}$ denote a set of potential facility locations and $N = \{1, 2, \dots, n\}$ denote a set of customers. In a single-source capacitated facility-location problem, each customer $j \in N$ has a demand a_j which must be satisfied by a single facility $i \in M$ with capacity b_i . The costs involved are a fixed cost, F_i , associated with each open facility i and a supply cost, c_{ij} , incurred when the demand of customer j is satisfied by facility i . The problem is to determine a subset of facilities to be opened and the customers assigned to each open facility at minimum total cost, and may be formulated as the following integer programme.

$$\text{P1:} \quad \text{Minimize} \left[\sum_{i \in M} F_i y_i + \sum_{i \in M} \sum_{j \in N} c_{ij} x_{ij} \right], \quad (1)$$

subject to

$$\sum_{j \in N} a_j x_{ij} \leq b_i \quad \forall i \in M, \quad (2)$$

$$\sum_{i \in M} x_{ij} = 1 \quad \forall j \in N, \quad (3)$$

$$x_{ij} \leq y_i \quad \forall i \in M, j \in N, \quad (4)$$

$$y_i \in \{0, 1\} \quad \forall i \in M, \quad (5)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in M, j \in N. \quad (6)$$

The interpretation of the decision variables is that $y_i = 1$ if facility i is open ($y_i = 0$ otherwise) and $x_{ij} = 1$ if facility i satisfies the demand of customer j ($x_{ij} = 0$ otherwise). Constraint set (2) ensures that the capacity of each facility is not exceeded. Constraint sets (3) and (4) ensure that the demand of each customer is satisfied by an open facility. The Lagrangian relaxation in which the capacity constraint set (2) is relaxed results in the following problem.

$$\text{P2:} \quad \text{Minimize} \left[\sum_{i \in M} F_i y_i + \sum_{i \in M} \sum_{j \in N} c_{ij} x_{ij} + \sum_{i \in M} \lambda_i \left(\sum_{j \in N} a_j x_{ij} - b_i \right) \right], \quad (7)$$

subject to (3)–(6).

Klincewicz and Luss propose solving the relaxed problem, for a given set of Lagrange multipliers $\{\lambda_i\}$, by using the dual ascent and dual adjustment procedures of Erlenkotter.²² A sequence of relaxed problems is solved whereby the Lagrange multipliers are updated using a version of subgradient optimization.

In the next section a class of single-source capacitated facility-location problems is determined for which the Lagrangian relaxation (P2) is guaranteed to produce an infeasible solution to the unrelaxed problem.

BOUNDS ON THE NUMBER OF OPEN FACILITIES

Lower bound on the number of open facilities in problem P1

Let A represent the total demand over all customers,

$$\left(A = \sum_{j=1}^n a_j \right).$$

Let $b_{(i)}$ be the capacity of the facility with the i th greatest capacity. Let L be a lower bound on the number of open facilities in a feasible solution to problem P1. L may be determined by relationship (8).

$$\sum_{i=1}^{L-1} b_{(i)} < A \leq \sum_{i=1}^L b_{(i)}. \quad (8)$$

Upper bound on the number of open facilities in problem P2

Consider a solution to problem P2 in which only two facilities are open. Label these facilities p and q . Let the set S_p contain the customers assigned to facility p and S_q those assigned to facility q ($S_p \cup S_q = N$ and $S_p \cap S_q = \emptyset$). The cost of this solution is given by (9):

$$\sum_{j \in S_p} c_{pj} + F_p + \lambda_p \sum_{j \in S_p} a_j + \sum_{j \in S_q} c_{qj} + F_q + \lambda_q \sum_{j \in S_q} a_j - \sum_{i \in M} \lambda_i b_i. \quad (9)$$

Without loss of generality, assume $\lambda_p \leq \lambda_q$.

Now consider a solution in which only facility p is open. All customers must be assigned to facility p , and the cost is given by (10):

$$\sum_{j \in N} c_{pj} + F_p + \lambda_p \sum_{j \in N} a_j - \sum_{i \in M} \lambda_i b_i. \quad (10)$$

The cost difference between the two solutions is shown as (11):

$$\sum_{j \in S_q} (c_{qj} - c_{pj}) + F_q + (\lambda_q - \lambda_p) \sum_{j \in S_q} a_j. \quad (11)$$

If condition (12) holds, the solution involving only p open has a lower cost than opening both p and q since (11) is guaranteed to be positive:

$$F_q > \sum_{j \in S_q} (c_{pj} - c_{qj}). \quad (12)$$

The condition depends on the allocation of customers to S_p and S_q . The most favourable treatment when both p and q are open is to allocate each customer to the facility with the lower supply cost, subject to each facility having at least one customer allocated to it. Under such treatment, condition (12) becomes that shown as (13), where the operator $\Sigma^{+(n-1)}$ denotes the summation of the $(n-1)$ most positive entities. If there are less than $(n-1)$ positive entities, then the operator denotes the summation of all positive entities:

$$F_q > \sum_{j \in N}^{+(n-1)} (c_{pj} - c_{qj}). \quad (13)$$

If condition (13) is satisfied for all $p, q \in M$, then there exists a solution to P2 with only one facility open which has a lower cost than any solution to P2 with two facilities open. We now wish to show, given that condition (13) holds for all $p, q \in M$, that the optimal solution to P2 will necessarily involve only one open facility.

Consider a solution to P2 involving k , ($k \geq 3$), open facilities: p , q and $(k-2)$ other facilities. Each open facility satisfies the demand of at least one customer. Reduce the problem by deleting the $(k-2)$ other facilities and their customers. Since condition (13) holds for the full problem, it will also hold for the reduced problem. Therefore a solution with p but not q open exists which has a lower cost than the solution with both p and q open. Thus a solution with $(k-1)$ facilities open can be obtained which has a lower cost than any solution with k facilities open. The argument can now be applied to a solution with $(k-1)$ facilities open to show that a better solution exists with only $(k-2)$ facilities open. By repeated application of the argument, it can be established that an optimal solution will involve only one open facility.

In cases where L , the lower bound on the number of open facilities [obtained from (8)] in problem P1, is greater than one and condition (13) holds for all $p, q \in M$, it is guaranteed that the optimal solution to P2 will not be feasible for P1 irrespective of the values assumed by the set of Lagrange multipliers.

It is interesting to note that, if a capacitated problem satisfies condition (13) for all $p, q \in M$, then the corresponding uncapacitated problem would necessarily have an optimal feasible solution involving only one open facility. In such cases the differences between the variable costs are

TABLE 1. Costs, customer demands and facility capacities

$i \backslash j$	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
c_{ij}	1	15	16	27	28	25	27	14	15	23	14	24	24	16	17	28	26	27	14	19
	2	20	13	15	24	13	16	15	15	20	18	16	16	12	17	10	23	17	24	22
	3	12	17	25	16	22	15	20	24	11	23	15	23	24	12	16	26	16	24	17
	4	25	27	16	23	21	25	26	24	26	13	15	27	18	18	23	27	23	25	23
	5	22	10	9	19	18	23	10	22	23	21	20	22	14	12	22	15	20	12	22
	6	20	16	24	19	26	24	19	17	20	22	27	14	24	16	26	27	16	16	13
	7	17	17	16	25	19	26	14	12	24	23	25	24	18	22	21	20	12	20	24
	8	18	20	23	22	28	18	19	17	15	20	24	20	19	24	19	16	18	25	19
	9	23	17	16	24	12	25	17	19	22	24	13	23	23	16	10	11	10	19	11
	10	14	14	16	20	25	12	23	23	18	27	15	21	27	21	16	15	25	12	26
a_j	17	24	16	16	20	15	20	18	18	20	10	21	19	23	15	19	15	10	20	11

TABLE 1 cont.

i	1	2	3	4	5	6	7	8	9	10
b_i	39	48	67	61	31	60	45	42	41	62
F_i	727	547	674	501	605	482	382	442	606	408

insufficient to warrant incurring additional fixed costs by opening more than one facility. It is difficult to judge the proportion of capacitated problems that fall into this class in practice but, clearly, a significant number of problems do not.

The optimal solution to problem P2 is guaranteed not to be feasible for problem P1 when the number of open facilities is less than L . Condition (13) can be tightened to show the condition under which the optimal solution to P2 is guaranteed to involve at most $(L - 1)$ open facilities. The maximum number of customers that can be allocated to any facility in a solution involving at least L open facilities is $(n - L + 1)$. Thus, if condition (14) holds for all $p, q \in N$, the optimal solution to problem P2 is guaranteed to involve at most $(L - 1)$ open facilities, and hence will be an infeasible solution to problem P1, irrespective of the values assumed by the Lagrange multipliers:

$$F_q > \sum_{j \in N}^{+(n-L+1)} (c_{pj} - c_{qj}). \tag{14}$$

Unfortunately, condition (14) is frequently no stronger in practice than condition (13), especially when L , the lower bound on the number of open facilities, is small relative to n , the number of customers.

EXAMPLE

To illustrate how the results of the previous section may be applied, a facility-location problem used by Bitran *et al.*²³ is considered. The problem involves 10 potential facility locations and 20 customers, and was originally a multiple-source problem whereby each customer may receive its supply from any number of facilities. In the example considered here, the problem is regarded as a single-source problem. Table 1 contains the fixed costs, F_i , the supply costs, c_{ij} , the facility capacities, b_i , and the customer demands, a_j , for $i \in M, j \in N$.

Total demand for the 20 customers (Σa_j) is 347. The total capacity of the six facilities with the greatest capacities is 343, while that of the seven facilities with the greatest capacities is 385. Application of relationship (8) shows that a lower bound on the number of open facilities in a feasible solution to problem P1 is seven.

Table 2 shows

$$\sum_{j \in N}^{+(n-1)} (c_{pj} - c_{qj})$$

for all $p, q \in M$, representing the right-hand side values of the inequality (13). The last row of Table 2 gives the values of $F_q^{(1)}, q \in M$, which is defined in (15):

$$F_q^{(1)} = \text{Max}_{p \in M} \left[\sum_{j \in N}^{+(n-1)} (c_{pj} - c_{qj}) \right]. \tag{15}$$

$F_q^{(1)}$ is the threshold value for F_q . If $F_q > F_q^{(1)}$ for all $q \in M$, then it is guaranteed that the optimal solution to problem P2 will involve only one open facility.

TABLE 2

		$\sum_{j \in N}^{+(n-1)}(c_{pj} - c_{qj})$									
$p \backslash q$		1	2	3	4	5	6	7	8	9	10
1	0	110	71	48	119	57	77	61	101	79	
2	27	0	57	14	43	28	27	19	48	26	
3	15	84	0	26	75	32	50	31	68	41	
4	63	112	97	0	105	82	77	78	111	83	
5	42	49	54	13	0	31	24	33	55	28	
6	38	92	69	48	89	0	60	42	88	71	
7	43	76	72	28	67	45	0	39	67	60	
8	42	83	68	44	91	42	54	0	94	63	
9	22	52	45	17	53	28	22	54	0	33	
10	46	76	64	35	72	57	61	49	79	0	
$F_q^{(1)}$	63	112	97	48	119	82	77	78	111	83	

By inspection it can be readily seen that F_q is considerably greater than $F_q^{(1)}$ for all $q \in M$. Thus the Lagrangian relaxation (P2) applied to this problem is guaranteed to result in only one open facility. Since the lower bound on the number of facilities in a feasible solution to problem P1 is seven, the optimal solution to P2 can never be a feasible solution to P1. Furthermore, in this problem, considering the relatively large fixed costs, the lower bound on the minimum cost for P1 from the solution to P2 is extremely poor.

CONCLUSIONS

The single-source capacitated facility-location problem and Lagrangian relaxation in which the capacity constraints are relaxed have been considered. Relationships between fixed costs and supply costs have been developed that identify problems for which the optimal solution to the relaxed problem is guaranteed not to be a feasible solution to the unrelaxed problem. This guarantee is unaffected by the values assumed by the Lagrange multipliers.

Lagrangian relaxation is generally regarded as a powerful and useful approach in solving hard combinatorial optimization problems. Fisher^{17,24} provides a review of the impressive track record achieved by the approach. However, as a result of investigating the single-source capacitated facility-location problem in which the capacity constraints are relaxed, it was found that Lagrangian relaxation is of no value for a defined subset of problem instances. Given a combinatorial optimization problem which can be partitioned into ‘easy’ and ‘hard’ constraints, it is tempting to use Lagrangian relaxation on the hard constraints. However, it may be worthwhile carrying out a preliminary analysis to determine the potential viability of the approach before extensive development takes place.

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